Statistical Foundations of Machine Learning

Slow-Paced Tutorial: Session 2

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Concept Review

Expectation

For a discrete random variable X,

$$\mathbb{E}[X] = \sum_{x \in S_X} x\, p_X(x).$$

For a continuous random variable X,

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x)\, dx.$$

More generally,

$$\mathbb{E}[g(X)] = \sum_{x \in S_X} g(x) p_X(x)$$

in the discrete case, and

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x)\, dx$$

in the continuous case.

Variance

If

$$\mu_X = \mathbb{E}[X],$$

then

$$\mathrm{Var}(X) = \mathbb{E}\left[(X - \mu_X)^2\right].$$

The useful computational formula is

$$\boxed{\mathrm{Var}(X) = \mathbb{E}[X^2] - \big(\mathbb{E}[X]\big)^2.}$$

Also,

$$\mathrm{Var}(aX + b) = a^2\, \mathrm{Var}(X).$$

If $X_1, \ldots, X_m$ are independent and have the same variance σ^2, then

$$\mathrm{Var}\left(\frac{1}{m}\sum_{i=1}^{m} X_i\right) = \frac{\sigma^2}{m}.$$

Important Distributions

Distribution	$\mathbb{E}[X]$	$\mathrm{Var}(X)$
$\mathrm{Ber}(p)$	p	$p(1-p)$
$\mathrm{Unif}(a,b)$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
$\mathcal{N}(\mu, \sigma^2)$	μ	σ^2
$\mathrm{Exp}(\lambda)$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$

Random Vector

A random vector (X, Y): The components may or may not be independent.

Joint CDF

The joint cumulative distribution function is

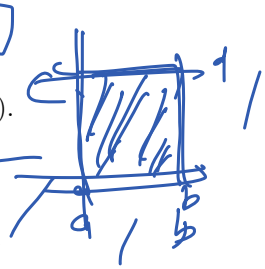
$$F_{XY}(x, y) = P(X \le x,\ Y \le y).$$

The marginal CDFs are

$$F_X(x) = \lim_{y \to \infty} F_{XY}(x, y), \qquad F_Y(y) = \lim_{x \to \infty} F_{XY}(x, y).$$

For $a < b$ and $c < d$,

$$P(a < X \le b,\ c < Y \le d) = F_{XY}(b, d) - F_{XY}(a, d) - F_{XY}(b, c) + F_{XY}(a, c).$$



Discrete Random Vectors

For discrete X and Y, the joint PMF is

$$p_{XY}(x, y) = P(X = x,\ Y = y).$$

It satisfies

$$\sum_x \sum_y p_{XY}(x, y) = 1.$$

The marginal PMFs are obtained by summing:

$$p_X(x) = \sum_y p_{XY}(x, y),$$

$$p_Y(y) = \sum_x p_{XY}(x, y).$$

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Conditional PMF

If $p_Y(y) > 0$, then

$$p_{X|Y}(x \mid y) = \frac{p_{XY}(x,y)}{p_Y(y)}.$$

For every fixed y,

$$\sum_x p_{X|Y}(x \mid y) = 1.$$

The law of total probability gives

$$p_X(x) = \sum_y p_{X|Y}(x \mid y) p_Y(y).$$

Independence

The random variables X and Y are independent if

$$F_{XY}(x,y) = F_X(x) F_Y(y)$$

for every x, y.

For discrete random variables,

X and Y are independent $\iff$ $p_{XY}(x,y) = p_X(x) p_Y(y)$.

Equivalently,

$$p_{X|Y}(x \mid y) = p_X(x), \qquad p_Y(y) > 0.$$

Jointly Continuous Variables

If (X, Y) has joint density f_{XY}, then

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x,y)\, dy,$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x,y)\, dx.$$

If $f_Y(y) > 0$, the conditional density is

$$f_{X|Y}(x \mid y) = \frac{f_{XY}(x,y)}{f_Y(y)}.$$

pmf

Joint pdf

For jointly continuous random variables,

X and Y are independent $\iff$ $f_{XY}(x,y) = f_X(x) f_Y(y)$

almost everywhere.

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Practice Questions

Q1. A fair die is thrown repeatedly until a 6 is obtained. Let X denote the number of throws required to obtain the first 6.

Find the PMF p_X and hence calculate $\mathbb{E}[X]$.

Q2. Let X be a continuous random variable with PDF

$$f_X(x) = \begin{cases} \dfrac{x}{2}, & 0 < x \leq 1, \\ \dfrac{1}{2}, & 1 < x \leq 2, \\ \dfrac{3-x}{2}, & 2 < x < 3, \\ 0, & \text{otherwise.} \end{cases}$$

Define

$$Y = X^2 - 5X + 3.$$

Without finding the PDF of Y, calculate $\mathbb{E}[Y]$.

Q3. A target consists of three concentric circles with radii

$$\frac{1}{\sqrt{3}}, \qquad 1, \qquad \sqrt{3}.$$

A shot earns 4 points if it falls inside the innermost circle, 3 points if it falls in the next ring, 2 points if it falls in the outermost ring, and 0 points if it falls outside the target.

Let X denote the distance of the hit from the centre. Suppose that

$$f_X(x) = \begin{cases} \dfrac{2}{\pi(1+x^2)}, & x > 0, \\ 0, & \text{otherwise.} \end{cases}$$

If S denotes the score obtained from one shot, find $\mathbb{E}[S]$.

Q4. A sample of size 3 is drawn with replacement from an urn containing 3 white balls, 2 black balls, and 3 red balls.

Let

$$X_1 = \text{number of white balls in the sample,}$$

and

$$X_2 = \text{number of black balls in the sample.}$$

Find the joint PMF $p_{X_1,X_2}(x_1, x_2)$ and the marginal PMFs p_{X_1} and p_{X_2}. Determine whether X_1 and X_2 are independent.

Q5. The joint PDF of (X, Y) is

$$f_{XY}(x, y) = \begin{cases} c\, e^{-(2x+3y)}, & 0 < x < y < \infty, \\ 0, & \text{otherwise,} \end{cases}$$

where c is a constant.

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(i) Find c.

(ii) Find the marginal PDFs f_X and f_Y.

(iii) Determine whether X and Y are independent.

(iv) Calculate

$$P\left(X < \frac{Y}{2}\right).$$

Q6. The joint PMF of (X, Y) is given by

$Y \backslash X$	1	2	3	4
4	0.08	0.11	0.09	0.03
5	0.04	0.12	0.21	0.05
6	0.09	0.06	0.08	0.04

(i) Find the marginal PMFs p_X and p_Y.

(ii) Find the conditional PMF $p_{X|Y}(x \mid 5)$.

(iii) Calculate

$$P(X + Y \leq 8) \qquad \text{and} \qquad P(X + Y > 7).$$

(iv) Calculate

$$P(XY \leq 14) \qquad \text{and} \qquad P(XY > 18).$$

(v) Find

$$P(X = 3 \mid Y = 5) \qquad \text{and} \qquad P(Y = 5 \mid X = 3).$$

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Q1. A fair die is thrown repeatedly until a 6 is obtained. Let X denote the number of throws required to obtain the first 6.

Find the PMF p_X and hence calculate $\mathbb{E}[X]$.

Fair die $\Rightarrow$ probability of getting 6 on any throw is $\boxed{p = \frac{1}{6}}$

probability of not getting a 6 is

$$q = 1 - \frac{1}{6} = \frac{5}{6}$$

$\boxed{p = \frac{1}{6}}$ $\boxed{q = \frac{5}{6}}$

Let X = # of throw required to obtain first 6

$$p_X(x) = P(X = x) = q^{x-1} p \qquad x = 1, 2, \ldots\ldots$$

$$E[X] = \sum_{x=1}^{\infty} x\, p_X(x) = \sum_{x=1}^{\infty} x\, q^{x-1} p$$

$$= p\,(1 + 2q + 3q^2 + \cdots)$$

denoting it by S

$$S = 1 + 2q + 3q^2 + \cdots$$

Subtract

multiply by q

$$qS = \quad q + 2q^2 + \cdots$$

$$(1-q)S = 1 + q + q^2 + \cdots$$

$$pS = \frac{1}{1-q} = \frac{1}{p}$$

$p = 1 - q$

$$\Rightarrow \quad \boxed{S = \frac{1}{p^2}}$$

$1 + r + r^2 + \cdots$, $|r| < 1$ $= \frac{1}{1-r}$

$$\therefore \quad E[X] = \frac{1}{p^2} \cdot p = \frac{1}{p} \quad \Rightarrow \quad \boxed{E[X] = 6}$$

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Q2. Let X be a continuous random variable with PDF

$$f_X(x) = \begin{cases} \frac{x}{2}, & 0 < x \le 1, \\ \frac{1}{2}, & 1 < x \le 2, \\ \frac{3-x}{2}, & 2 < x < 3, \\ 0, & \text{otherwise.} \end{cases}$$

Define

$$Y = X^2 - 5X + 3$$

Without finding the PDF of Y, calculate $\mathbb{E}[Y]$.

Integration Formula

$$\int x^n dx = \frac{x^{n+1}}{n+1} + c$$

$$E[Y] = \int y\, f_Y(y)\, dy$$

$$Y = g(X) = X^2 - 5X + 3 \rightarrow E[Y] = E[X^2] - 5E[X] + 3$$

$$E[Y] = E(g(X)) = \int_{-\infty}^{\infty} g(x)\, f_X(x)\, dx$$

$$= \int_{-\infty}^{\infty} (x^2 - 5x + 3)\, f_X(x)\, dx$$

$$= \int_0^1 \underbrace{(x^2-5x+3)}_{g(x)} \frac{x}{2} dx + \int_1^2 \underbrace{(x^2-5x+3)}_{g(x)} \frac{1}{2} dx + \int_2^3 \underbrace{(x^2-5x+3)}_{g(x)} \frac{3-x}{2} dx$$

$$= \frac{1}{2}\int_0^1 (x^3 - 5x^2 + 3x)\, dx + \frac{1}{2}\int_1^2 (x^2 - 5x + 3)\, dx + \frac{3}{2}\int_2^3 (x^2 - 5x + 3)\, dx - \frac{1}{2}\int_2^3 (x^3 - 5x^2 + 3x)\, dx$$

$$= \frac{1}{2}\left[\frac{x^4}{4} - \frac{5x^3}{3} + \frac{3x^2}{2}\right]_0^1 + \frac{1}{2}\left[\frac{x^3}{3} - \frac{5x^2}{2} + 3x\right]_1^2 + \frac{3}{2}\left[\frac{x^3}{3} - \frac{5x^2}{2} + 3x\right]_2^3 - \frac{1}{2}\left[\frac{x^4}{4} - \frac{5x^3}{3} + \frac{3x^2}{2}\right]_2^3$$

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$$= \frac{1}{2}\left(\frac{1}{4}-\frac{5}{3}+\frac{3}{2}\right)+\frac{1}{2}\left(\frac{8}{3}-\frac{20}{2}+6-\frac{1}{3}+\frac{5}{2}-3\right)+\frac{3}{2}\left(\frac{27}{3}-\frac{45}{2}+9-\frac{8}{3}+\frac{20}{2}-6\right)$$
$$-\frac{1}{2}\left(\frac{81}{4}-\frac{135}{3}+\frac{27}{2}-\frac{16}{4}+\frac{40}{3}-\frac{12}{2}\right)$$

$$= \frac{1}{2}\left(\frac{1}{12}\right)+\frac{1}{2}\left(\frac{-13}{6}\right)+\frac{3}{2}\left(\frac{-19}{6}\right)-\frac{1}{2}\left(\frac{-95}{12}\right)$$

$$= \frac{1}{24}-\frac{13}{12}-\frac{57}{12}+\frac{95}{24} = -\frac{44}{24} = \boxed{-\frac{11}{6}}$$

You can compute by following techniques also

$$E[X] = \int_{-\infty}^{\infty} x\, f_X(x)\,dx = \int_0^1 x\cdot\frac{x}{2}\,dx + \int_1^2 x\cdot\frac{1}{2}\,dx + \int_2^3 x\cdot\frac{3-x}{2}\,dx$$

$$= \frac{1}{2}\left[\frac{x^3}{3}\right]_0^1 + \frac{1}{2}\left[\frac{x^2}{2}\right]_1^2 + \frac{1}{2}\left[\frac{3x^2}{2}-\frac{x^3}{3}\right]_2^3$$

$$= \frac{3}{2}$$

$$E[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x)\,dx = \int_0^1 x^2\cdot\frac{x}{2}\,dx + \int_1^2 x^2\cdot\frac{1}{2}\,dx + \int_2^3 x^2\frac{3-x}{2}\,dx$$

$$= \frac{1}{2}\int_0^1 x^3\,dx + \frac{1}{2}\int_1^2 x^2\,dx + \frac{1}{2}\int_2^3 (3x^2-x^3)\,dx$$

$$= \frac{1}{2}\left[\frac{x^4}{4}\right]_0^1 + \frac{1}{2}\left[\frac{x^3}{3}\right]_1^2 + \frac{1}{2}\left(x^3-\frac{x^4}{4}\right)_2^3$$

$$= \boxed{\frac{8}{3}}$$

$$E[Y] = E[X^2] - 5E[X] + 3$$

$$= \frac{8}{3} - 5\cdot\frac{3}{2} + 3$$

$$= \boxed{\frac{-11}{6}}$$

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Let X denote the distance of the hit from the centre. Suppose that

$$f_X(x) = \begin{cases} \dfrac{2}{\pi(1+x^2)}, & x > 0, \\ 0, & \text{otherwise.} \end{cases}$$

If S denotes the score obtained from one shot, find $\mathbb{E}[S]$.

Integration Formula

$$\int \frac{1}{1+x^2}\,dx = \tan^{-1}x$$

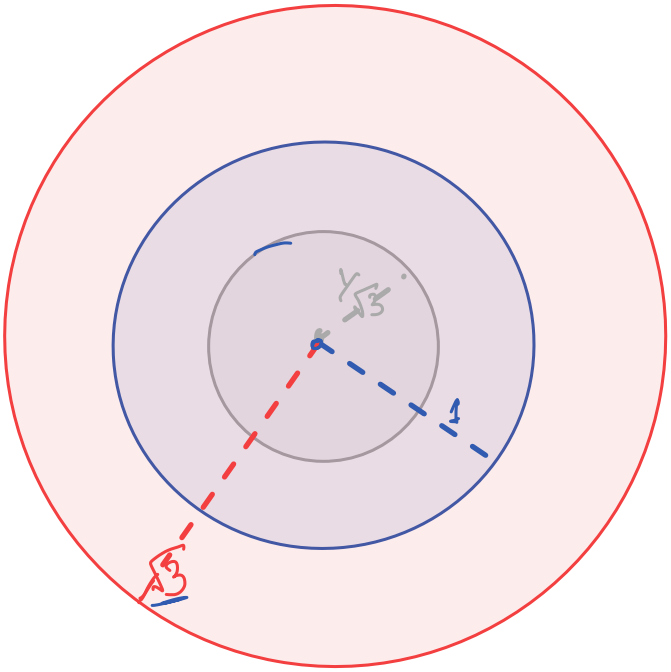
$\tan^{-1}0 = 0$

$\tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$

$\tan^{-1}(1) = \frac{\pi}{4}$

$\tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$

$\tan^{-1}(-x) = -\tan^{-1}x$



Let S denotes the score obtained

$$S = \begin{cases} 4 & ; \ 0 < X \le \frac{1}{\sqrt{3}} \\ 3 & ; \ \frac{1}{\sqrt{3}} < X \le 1 \\ 2 & ; \ 1 < X \le \sqrt{3} \\ 0 & ; \ X > \sqrt{3} \end{cases}$$

$$\int \frac{1}{1+x^2}\,dx = \tan^{-1}x$$

CDF of X

$$F_X(x) = P(X \le x) = \int_0^x \frac{2}{\pi(1+t^2)}\,dt$$

$$= \frac{2}{\pi}\left(\tan^{-1}t\right)_0^x = \boxed{\frac{2}{\pi}\tan^{-1}x}$$

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$$P(S=4) = P\left(0 < X \le \frac{1}{\sqrt{3}}\right) = \frac{2}{\pi}\left(\tan^{-1}\frac{1}{\sqrt{3}} - \tan^{-1}0\right)$$

$$= F_X\left(\frac{1}{\sqrt{3}}\right) - F_X(0) = \frac{2}{\pi}\cdot\frac{\pi}{6} = \frac{1}{3}$$

$$P(S=3) = P\left(\frac{1}{\sqrt{3}} < X \le 1\right) = \frac{2}{\pi}\left[\tan^{-1}1 - \tan^{-1}\frac{1}{\sqrt{3}}\right]$$

$$= \frac{2}{\pi}\left[\frac{\pi}{4} - \frac{\pi}{6}\right] = \frac{1}{6}$$

$$P(S=2) = P\left(1 < X \le \sqrt{3}\right) = \frac{2}{\pi}\left(\tan^{-1}\sqrt{3} - \tan^{-1}1\right)$$

$$= \frac{2}{\pi}\left[\frac{\pi}{3} - \frac{\pi}{4}\right] = \frac{1}{6}$$

$$E[S] = 4\,P\left(0 < X \le \frac{1}{\sqrt{3}}\right) + 3\,P\left(\frac{1}{\sqrt{3}} < X \le 1\right) + 2\,P\left(1 < X \le \sqrt{3}\right)$$

$$= 4\cdot\frac{1}{3} + 3\cdot\frac{1}{6} + 2\cdot\frac{1}{6}$$

$$= \frac{13}{6}$$

$$\boxed{E[S] = \frac{13}{6}}$$

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Q4. A sample of size 3 is drawn with replacement from an urn containing 3 white balls, 2 black balls, and 3 red balls.

Let

$$X_1 = \text{number of white balls in the sample,}$$

and

$$X_2 = \text{number of black balls in the sample.}$$

Find the joint PMF $p_{X_1,X_2}(x_1, x_2)$ and the marginal PMFs p_{X_1} and p_{X_2}. Determine whether X_1 and X_2 are independent.

(X_1, X_2)

3W 2B 3R $\longrightarrow$ Total = 8 balls

$P(W) = \frac{3}{8}$ $P(B) = \frac{2}{8} = \frac{1}{4}$ $P(R) = \frac{3}{8}$

If $X_1 = x_1$, $X_2 = x_2$ (X_1, X_2)

then X_3 = number of red balls $= 3 - x_1 - x_2$

$x_1 \geq 0$ $x_2 \geq 0$ $x_1 + x_2 \leq 3$

$3 - x_1 - x_2 \geq 0$

$x_1 + x_2 \leq 3$

$$\therefore p_{X_1 X_2}(x_1, x_2) = P(X_1 = x_1, X_2 = x_2)$$

$$= \binom{3}{x_1}\binom{3-x_1}{x_2}\left(\frac{3}{8}\right)^{x_1}\left(\frac{2}{8}\right)^{x_2}\left(\frac{3}{8}\right)^{3-x_1-x_2}$$

$$\binom{n}{r} = \frac{n!}{r!\,(n-r)!}$$

$$= \frac{3!}{x_1!\,x_2!\,(3-x_1-x_2)!}\left(\frac{3}{8}\right)^{x_1}\left(\frac{2}{8}\right)^{x_2}\left(\frac{3}{8}\right)^{3-x_1-x_2}$$

Marginal of X_1

$$p_{X_1}(x_1) = \sum_{\text{over } x_2} p_{X_1 X_2}(x_1, x_2)$$

3W

$$X_1 \sim \text{Bino}\left(3, \frac{3}{8}\right)$$

$$p_{X_1}(x) = \binom{3}{x}\left(\frac{3}{8}\right)^{x}\left(\frac{5}{8}\right)^{3-x}$$

p $(1-p)$

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Marginal of X_2

$n = 3$, $\frac{1}{4}$

$$X_2 \sim \text{Bino}\left(3, \frac{1}{4}\right)$$

$$P_{X_2}(x) = \binom{3}{x} \underbrace{\left(\frac{1}{4}\right)^{x}}_{p} \underbrace{\left(\frac{3}{4}\right)^{3-x}}_{(1-p)}$$

Independence ?

Check

$$P(X_1 = x_1, X_2 = x_2) = P(X_1 = x_1)\, P(X_2 = x_2) \quad \forall x_1, x_2$$

Since $P(X_1 = 3, X_2 = 3) = 0$ $\quad x_1 + x_2 \leq 3$

and $P(X_1 = 3) > 0$ $\quad P(X_2 = 3) > 0$

$$\Rightarrow \quad P(X_1 = 3, X_2 = 3) \neq P(X_1 = 3)\, P(X_2 = 3)$$

Hence Not independent.

Also check

$$P(X_1 = 0, X_2 = 0) = P(X_1 = 0)\, P(X_2 = 0) \ ?$$

Not equal

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Q5. The joint PDF of (X, Y) is

$$f_{XY}(x,y) = \begin{cases} c\,e^{-(2x+3y)}, & 0 < x < y < \infty, \\ 0, & \text{otherwise}, \end{cases}$$

where c is a constant.

(i) Find c.

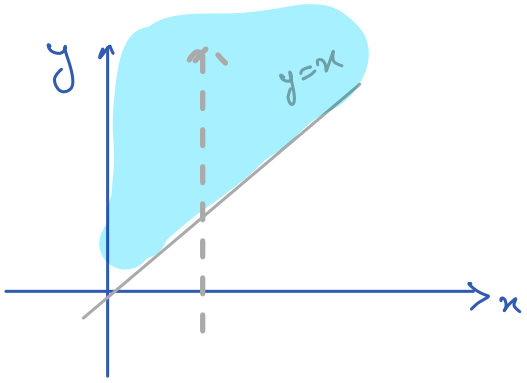
(ii) Find the marginal PDFs f_X and f_Y.

(iii) Determine whether X and Y are independent.

(iv) Calculate

$$P\left(X < \frac{Y}{2}\right).$$

i) To check valid pdf $\iint f_{XY}(x,y)\,dx\,dy = 1$ and $f_{XY}(x,y) \geq 0$



$$\int_{x=0}^{\infty}\int_{y=x}^{\infty} c\,e^{-(2x+3y)}$$

$$= \int_{x=0}^{\infty} c\,e^{-2x}\left(\int_{y=x}^{\infty} e^{-3y}\,dy\right)dx$$

$$= c\int_{x=0}^{\infty} e^{-2x}\left[\frac{e^{-3y}}{-3}\right]_{x}^{\infty} dx$$

$$= \frac{c}{3}\int_{0}^{\infty} e^{-2x}\left(0 + \frac{e^{-3x}}{3}\right)dx$$

$$= \frac{c}{3}\int_{0}^{\infty} e^{-5x}\,dx$$

$$= \frac{c}{3}\left(\frac{e^{-5x}}{-5}\right)_{0}^{\infty}$$

$$= \frac{c}{3}\left(0 + \frac{1}{5}\right) = \frac{c}{15}$$ ← this should be equal to 1

$$\Rightarrow \boxed{c = 15}$$

ii) For $x > 0$ $\quad f_X(x) = \int_{x}^{\infty} 15\,e^{-(2x+3y)}\,dy = 5e^{-5x}$

For $y > 0$ $\quad f_Y(y) = \int_{0}^{y} 15\,e^{-(2x+3y)}\,dy = \frac{15}{2}\left[e^{-3y} - e^{-5y}\right]$

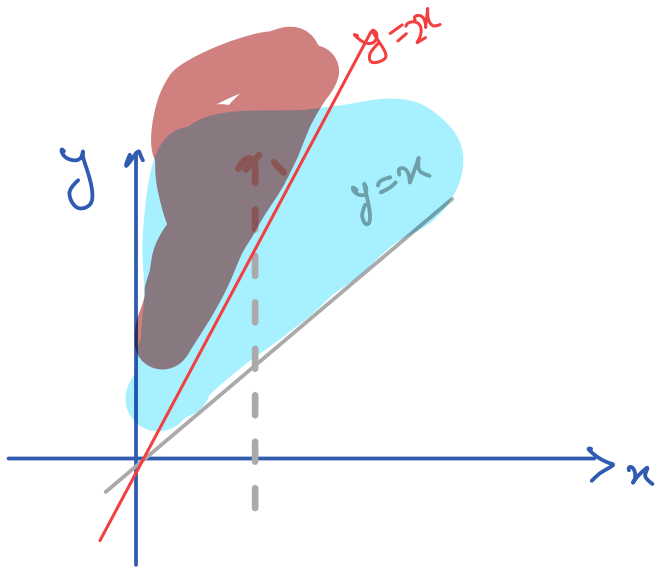
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iii) Since support $0<x<y$ is triangular and not the product of the two marginal support.

therefore

X and Y are not independent.

iv) $P\left(X<\frac{Y}{2}\right) = \int_{x=0}^{\infty}\int_{2x}^{\infty} 15\, e^{-(2x+3y)}\, dy\, dx$



$= 5\int_{0}^{\infty} e^{-8x}\, dx$

$= 5/8$

$x<\frac{y}{2} \Rightarrow y>2x$

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Q6. The joint PMF of (X, Y) is given by

$Y \backslash X$	1	2	3	4
4	0.08	0.11	0.09	0.03
5	0.04	0.12	0.21	0.05
6	0.09	0.06	0.08	0.04

(i) Find the marginal PMFs p_X and p_Y.

(ii) Find the conditional PMF $p_{X|Y}(x \mid 5)$.

(iii) Calculate

$$P(X+Y \leq 8) \quad \text{and} \quad P(X+Y > 7).$$

(iv) Calculate

$$P(XY \leq 14) \quad \text{and} \quad P(XY > 18).$$

(v) Find

$$P(X = 3 \mid Y = 5) \quad \text{and} \quad P(Y = 5 \mid X = 3).$$

i) Marginals

x	1	2	3	4
$p_X(x)$	0.21	0.29	0.38	0.12

By adding the columns

y	4	5	6
$p_Y(y)$	0.31	0.42	0.27

By adding rows

ii) Conditionals PMF $p_{X|Y}(x|5)$

$$p_{X|Y}(x|5) = \frac{p_{XY}(x,5)}{p_Y(5)}$$

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$$P_{X|Y}(1|5) = \frac{P_{XY}(1,5)}{P_Y(5)} = \frac{0.04}{0.42} = \frac{2}{21}$$

$$P_{X|Y}(2|5) = \frac{P_{XY}(2,5)}{P_Y(5)} = \frac{2}{7}$$

$$P_{X|Y}(3|5) = \frac{P_{XY}(3,5)}{P_Y(5)} = \frac{1}{2}$$

$$P_{X|Y}(4|5) = \frac{P_{XY}(4,5)}{P_Y(5)} = \frac{5}{42}$$

iii) $P(X+Y \leq 8)$

Ans: 0.83

$Y \backslash X$	1	2	3	4
4	0.08	0.11	0.09	0.03
5	0.04	0.12	0.21	0.05
6	0.09	0.06	0.08	0.04

Just add these value

$P(X+Y > 7)$

0.47

$Y \backslash X$	1	2	3	4
4	0.08	0.11	0.09	0.03
5	0.04	0.12	0.21	0.05
6	0.09	0.06	0.08	0.04

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(iii) $P(XY \leq 14) = 0.59$

$Y \backslash X$	1	2	3	4
4	0.08	0.11	0.09	0.03
5	0.04	0.12	0.21	0.05
6	0.09	0.06	0.08	0.04

$P(XY > 18) = 0.09$

$Y \backslash X$	1	2	3	4
4	0.08	0.11	0.09	0.03
5	0.04	0.12	0.21	0.05
6	0.09	0.06	0.08	0.04

(v) Find

$$P(X = 3 \mid Y = 5) \quad \text{and} \quad P(Y = 5 \mid X = 3).$$

$$P_{X|Y}(3|5) = \frac{P_{XY}(3,5)}{P_Y(5)} = \boxed{\frac{1}{2}}$$

$$P_{Y|X}(5|3) = \frac{P_{X,Y}(3,5)}{P_X(3)} = \boxed{\frac{21}{38}}$$

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